

**Started on** Friday, 21 August 2026, 4:06 PM**State** Finished**Completed on** Friday, 21 August 2026, 4:16 PM**Time taken** 9 mins 36 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6      int count=1;
7      int c = 0;
8      for(int i=n/2; i<n; i++)
9      {
10         count++;
11         for(int j=1; j<n; j = 2 * j)
12         {
13             count++;
14             for(int k=1; k<n; k = k * 2)
15             {
16                 count++;
17                 c++;
18                 count++;
19             }
20             count++;
21         }
22         count++;
23     }
24     count++;
25     printf("%d",count);
26 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)